CS4.406 Term Project — Mid-Submission Report

How fast does a news recommender go stale?

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1 What we are studying

A news recommender learns from past clicks and articles, then stays frozen. But news articles are replaced within hours. We want to know how fast a frozen recommender gets worse, and which kind of signal gets worse first: **word matching** (BM25), **meaning matching** (a frozen multilingual MiniLM model), or **click history** (which articles were read by the same people). Recency and popularity are simple baselines.

How we measure it. Take a real click. List the articles published in the 48 hours before it. Rank the clicked article among them (metrics: MRR and recall@10, with bootstrap confidence intervals). Do this for models frozen on different days and compare each one with a model rebuilt the day before. The gap is the *decay*.

2 Data

	EB-NeRD (main)	**NPR** (second)	**Adressa** (optional)
Publisher, language	Ekstra Bladet, Danish	Globo, Portuguese	Norwegian news site
Days of clicks	22 (May 2023)	118 (Jan–Apr 2023)	90 (Jan–Mar 2017)
Clicks	113.4 M	17.6 M	35.2 M
Users	791,582	1,162,802	3,614,911
Typical age of a clicked article	1.4 h	13.8 h	5.8 h

All three were cleaned into small per-day click files inside the project, and the article text was embedded on the GPU. For EB-NeRD we use the click history of the validation users, because it is the one part whose volume and type of click stay the same across dates.

3 Pipeline

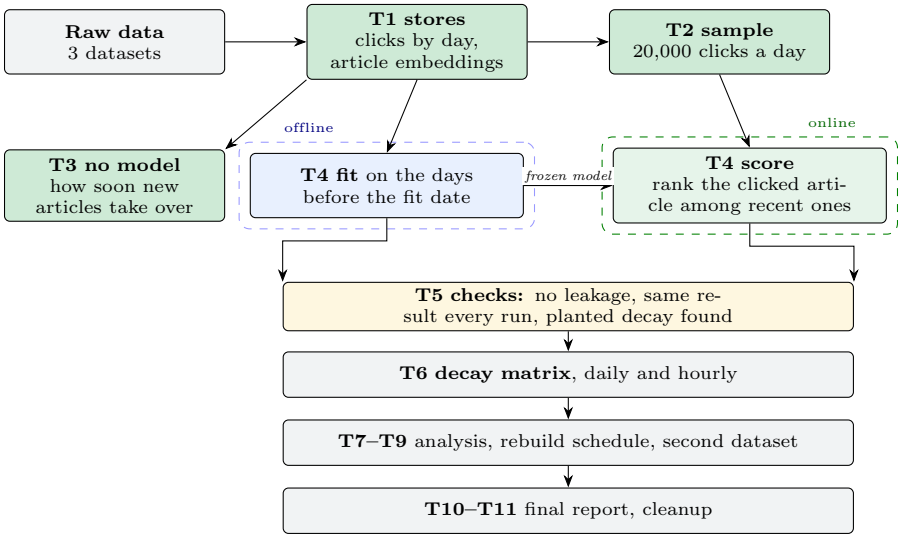


Figure 1: The pipeline. Green is built and tested, yellow is next, grey is still to do. A monitor watches time, memory and disk in every step.

4 What is done

Stages T1 to T4 are built, and 30 automated tests pass. In short:

- The three click logs are cleaned and stored, with duplicates removed and counted.
- Five scorers work: word matching, meaning matching, click history, recency and popularity, plus a random shuffle to compare against.
- The sampling gives the same files with 1, 3 or 7 workers. Three deliberate leak bugs were each caught by a test.
- **First measurement, no model needed:** after a model is frozen, how many new clicks go to articles it never saw? At 6 hours it is **90.8%** [88.5, 92.5] for EB-NeRD, **48.5%** [45.4, 51.5] for Adressa and **15.5%** [11.3, 20.1] for NPR (95% bootstrap intervals over fit days). So news turns over very differently across publishers.
- **Pipeline check, not a result:** on one test case (fit 16 May, day 18 May, 19,939 clicks), recency scored MRR 0.177 [0.174, 0.181] and the random shuffle 0.015 [0.014, 0.016]. This shows the scoring runs; it is not a finding about decay.

5 What went wrong, and what we changed

- Our first plan assumed one uniform click corpus; the data is not uniform, so we chose a single uniform corpus (click-forward evaluation).
- The click-history model needed 7.1 GB, not the 1.5 GB we planned, so only 2 can run at a time.
- The first stores had duplicate rows (107,633 in NPR, 7,693 in Adressa); we found them by counting and rebuilt.

6 Remaining work

Stage	What it will produce
T5	The checks that must pass before any result is trusted: no leakage, repeatable runs, a planted decay that is found
T6	The decay matrix on EB-NeRD: 14 fit dates, about 105 day-by-day cells, with fixed and live user profiles; plus an hourly version
T7	Which signal goes stale first, with confidence intervals
T8, T9	How often a rebuild pays off; the same study on NPR, then Adressa
T10, T11	Final report, AI log, cleanup

Still to decide: whether the headline metric counts clicks on articles the model could not see, and who owns which stage. No decay result is claimed yet.